INCH-POUND

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w/AMENDMENT 2
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SUPERSEDING
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w/AMENDMENT 1
12 April 1991

MILITARY SPECIFICATION

BRUSHES, CLEANING, FOR BORE, CHAMBER AND FIRING PIN HOLE OF SMALL ARMS

Inactive for new design after 11 August 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers bore, chamber and firing pin hole cleaning brushes for small arms.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	-Preservation, Methods of.
MIL-W-13855	-Weapons, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-14232	-Parts, Equipment and Tools for Army Materiel, Packaging and Packing of.

STANDARDS

Military

MIL-STD-105	-Sampling Procedures and Tables for Inspection by Attributes.

FSC 1005

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MIL-STD-109 -Quality Assurance Terms and Definitions

(Copies of these documents are available online at http://quicksearch.dla.mil/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5904.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified (see 6.2), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. Brushes shall conform to the materials, construction and design requirements in accordance with the applicable drawings specified in the contract (see 6.2) and MIL-W-13855.

3.2.1 Bristle weight. The bristle weight shall be in accordance with the applicable drawings specified in the contract (see 6.2).

3.2.2 Bristle distribution. Unless otherwise specified on the drawings, bristles shall be evenly distributed.

3.2.3 Protective finish. The protective finish shall be in accordance with the applicable drawings specified in the contract (see 6.2).

3.2.4 Nail load. The nail load, for ratchet type brushes only, shall be in accordance with the applicable drawings specified in the contract (see 6.2).

3.2.5 Tensile Pull. The tensile pull, as applicable, shall be in accordance with the applicable drawings specified in the contract (see 6.2).

3.2.6 Ratchet type brushes. Ratchet type brushes, after final assembly, shall have freely rotating ratchet members.

3.3 Performance.

3.3.1 Bore cleaning brushes. Brushes shall withstand 1000 strokes through the full length of the type and caliber barrel specified. After 1000 strokes, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

3.3.2 Chamber cleaning brushes. Brushes shall withstand 1000 insertions into the full depth of the applicable chamber and while fully inserted on each insertion, the brush shall be rotated one (1) full revolution. Speed of revolution shall be 15 to 30 revolutions per minute.

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After 1000 revolutions and insertions, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

3.3.3 Firing pin hole cleaning brush. Brushes shall withstand 1000 strokes into the specified firing pin hole of the applicable bolt. After 1000 strokes, the bristles shall not be loose in the core wire and bristle weight shall not be less than 75 percent of the original weight.

3.4 Workmanship. Cleaning brushes shall be in accordance with the workmanship requirements of MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of Inspection. The inspection requirements specified herein are classified as follows:

a) First article inspection (see 4.4).
b) Quality conformance inspection (see 4.5).

4.4 First article inspection. A first article shall be selected from early production and submitted for testing in accordance with contract requirements. The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be no larger than one week's production and within the limitations of MIL-W-13855.

4.5.2 Sampling.

4.5.2.1 Sampling for examination of brushes shall be in accordance with MIL-STD-105 inspection level I.

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4.5.2.2 Sampling for packaging shall be in accordance with MIL-P-116.

4.5.2.3 Sampling for packing shall be in accordance with MIL-STD-105 using inspection level I.

4.5.2.4 Sampling for the bristle weight test shall be in accordance with inspection level S2 of MIL-STD-105. Acceptance shall be based on an AQL of 1.0 percent defective.

4.5.2.5 The sample for the performance, nail load and tensile pull tests as applicable shall be ten brushes randomly selected from each lot.

4.5.3. Examination

4.5.3.1 Packaging. Examination of packaging shall be as prescribed for group A and B of MIL-P-116.

4.5.3.2 Packing. Examination of packing shall be in accordance with MIL-P-14232.

4.5.4 Testing.

4.5.4.1 Government furnished property. Barrels and bolts to be used for the applicable performance tests shall be of the type and caliber specified on the applicable brush drawing. Each barrel and bolt shall be clean and dry and no barrel or bolt shall be used for more than 500 brush tests.

4.5.4.2 Bristle weight. Brushed selected in accordance with 4.5.2.4 shall be tested in accordance with 4.6.1 for compliance with the bristle weight specified on the applicable drawing.

4.5.4.3 Performance-bore cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.2 for compliance with 3.3.1. Failure of any brush shall be cause for rejections of the represented lot.

4.5.4.4 Performance-chamber cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.3 for compliance with 3.3.2. Failure of any brush shall be cause for rejection of the represented lot.

4.5.4.5 Performance-firing pin hole cleaning brushes. Brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.4 for compliance with 3.3.3. Failure of any brush shall be cause for rejection of the represented lot.

4.5.4.6 Nail load. When applicable, brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.5 for compliance with the nail load requirements specified on the applicable drawing. Failure of any brush shall be cause for rejection of the represented lot.

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4.5.4.7 Tensile pull. When applicable, brushes selected in accordance with 4.5.2.5 shall be tested in accordance with 4.6.6 for compliance with the tensile pull requirements specified on the applicable drawing. Failure of any brush shall be cause for rejection of the represented lot.

4.6 Test Methods.

4.6.1 Bristle weight. The bristles of each brush shall be removed from the brush and weighed. The brush shall have failed when the bristle weight is not within the limits of the bristle weight specified in the applicable brush drawing.

4.6.2 Performance-bore cleaning brushes. The brush shall be pushed in one continuous motion through the length of the applicable barrel and returned. This movement shall be counted as two strokes. The brush shall be forced entirely clear of the barrel at the end of each stroke. Speed of travel for each test brush shall be approximately 2 feet per second. The number of strokes required for each test brush shall be specified in 3.3.1. Barrels used for this test shall be of the type and caliber specified on the applicable brush drawing. Upon completion of this test the weight loss of the bristles shall be determined. The weight loss of the bristles shall be the original weight of the brush minus the weight of the brush after the test.

4.6.3 Performance-chamber cleaning brushes. The brush shall be inserted into the full depth of the chamber, rotated clockwise one (1) revolution and removed the number of times specified in 3.3.2. A 360^0 movement shall be considered as one revolution. Speed of revolution shall be 15 to 30 revolutions per minute. Chambers used for this test shall be of the type and caliber specified on the applicable brush drawing. Upon completion of this test, the weight loss of the bristles shall be determined as specified in 4.6.2.

4.6.4 Performance-firing pin hole cleaning brushes. The bristle portion of the brush shall be pushed in one continuous motion through the length of the firing-pin hole of the applicable bolt and returned. This movement shall be counted as two strokes. The bristle portion of the brush shall be forced entirely clear of the firing-pin hole at the end of each stroke. Speed of travel for each stroke shall be approximately 2 feet per second. The number of strokes required for each test brush shall be as specified in 3.3.3. Bolts used for this test shall be of the type and caliber specified. Upon completion of this test the weight loss of the bristles shall be determined as specified in 4.6.2.

4.6.5 Nail load. The brush shall be assembled into the Government approved test gage and subjected to the nail load test. The brush is defective is the applied load disengages the nail completely from the adapter.

4.6.6 Tensile pull. The brush shall be assembled into the Government approved test gage and subjected to the tensile pull test. The brush is defective if the brush does not meet the requirements specified on the applicable drawing.

4.7 Inspection equipment. Inspection equipment utilized in quality conformance inspection shall be as listed on the applicable Inspection Equipment List (see 6.2).

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5. PREPARATION FOR DELIVERY

5.1 Pilot pack. The pilot pack shall consist of a brush or brushes as required by and packaged in accordance with the applicable packaging datasheet for the level of protection specified in the contract (see 6.2), packed level C and forwarded in accordance with 3.1.

5.2 Preservation, packaging, packing and marking. Brushes shall be preserved, packaged, packed and marked in accordance with the applicable packaging data sheet for the level of protection specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. This specification is intended to cover, but not be limited to, procurement of the following brushes:

Drawing		Nomenclature
C5504036	-	Brush, Cleaning, Caliber .45, M5
C5504037	-	Brush, Cleaning, Caliber .50, M4, Bore
C5564174	-	Brush, Cleaning, Caliber .30
C5564179	-	Brush, Cleaning, Caliber .22, M3
D6528362	-	Brush, Chamber Cleaning, M1
C7162132	-	Brush, Cleaning, Caliber .38
B7162702	-	Brush, Cleaning, Firing Pin Hole, Caliber .50
B7225087	-	Brush, Bore, M25, 20MM
B7790452	-	Brush, Chamber Cleaning, 7.62MM
B7790582	-	Brush, Chamber Cleaning, Caliber .30
B7790617	-	Brush, 40MM
B7790655	-	Brush Assembly, 40MM
B7790737	-	Brush, Cleaning, Small Arms: Chamber (Caliber .50)
B7791063	-	Brush, Chamber Cleaning, 20MM
C8407954	-	Brush, Chamber Cleaning (Caliber .50 – M6 and M3)
B11010305	-	Brush Assembly, 40 MM

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification
b. List of applicable drawings and specifications pertinent to the brushes, showing applicable revision dates.
c. Requirements for submission of first article and shipping instructions for first article and pilot packages (see 3.1, 4.4 and 5.1).
d. Applicable packaging data sheet (see 6.3).
e. Selection of applicable levels of packaging and packing.
f. Master Lists of Inspection and Test Equipment (see 4.7).
g. Government property to be furnished to the contractor and responsibilities therefore (see 4.5.4.1).

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6.3 The following packaging data sheets are applicable to the brushes covered by this specification as specified in 6.1.

P5504036 - Packaging Data Sheet, Brush, Cleaning, Caliber .45, M5
P5504037 - Packaging Data Sheet, Brush, Cleaning, Caliber .50, M4, Bore
P5564174 - Packaging Data Sheet, Brush, Cleaning, Caliber .30
P5564179 - Packaging Data Sheet, Brush, Cleaning, Caliber .22, M3
P6528362 - Packaging Data Sheet, Brush, Chamber Cleaning, M1
P7162132 - Packaging Data Sheet, Brush, Caliber .38
P7162702 - Packaging Data Sheet, Brush, Cleaning, Firing Pin Hole, Caliber .50
P7225087 - Packaging Data Sheet, Brush, Bore, M25, 20MM
P7790452 - Packaging Data Sheet, Brush, Chamber Cleaning, 7.62MM
P7790582 - Packaging Data Sheet, Brush, Chamber Cleaning, Caliber .30
P7790665 - Packaging Data Sheet, Brush Assembly, 40MM
P7790737 - Packaging Data Sheet, Brush, Cleaning, Small Arms, Chamber (Caliber 0.5)
P11010305 - Packaging Data Sheet, Brush Assembly, 40MM
P7791063 - Packaging Data Sheet, Brush, Chamber Cleaning, 20MM
P8407954 - Packaging Data Sheet, Brush, Chamber Cleaning, (Caliber .50 – M6 and M3)

6.4 Management control system requirements. When warranted, the contract should specify the application of MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.5 Performance tests. The Government installation or activity will conduct the quality conformance performance test when it is not feasible to furnish the contractor the barrels or bolts needed for the tests.

6.6 The margins of this specification are marked with an asterisk to indicate where changes (addition, modifications, corrections, deletions,) from the previous issue were made. This was done as a convenience only and the Governments assume no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Custodians:

Army –AR

Air Force – 84

Navy – OS

Preparing Activity:

Army – AR

Project (1005-2015-008)

Review Activities:

Navy - MC

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